CS4928 Project P32: Downstream Maintainability of AI-Assisted Code

Primary | C. AI-Assisted Software Design & Quality

Item	Details
Assigned study	Borg et al., "Echoes of AI: Investigating the downstream effects of AI assistants on software maintainability"
Venue / year	Empirical Software Engineering (2026)
Language route	Java direct
Expected setup	Low-Medium

Paper: https://link.springer.com/article/10.1007/s10664-026-10889-1

Authors' complete replication repository: https://github.com/codescene-research/echoes-of-ai-emse-2025

Your bounded starting task

Reproduce one bounded result from the public experiment data comparing subsequent evolution of AI-assisted versus non-AI solutions. Audit the maintainability measure and inspect a small set of code solutions qualitatively.

Sampling/freeze rule: before the main M2 analysis, record the exact case/project/file identifiers used. Unless the bounded task names a different exact set, take the first 12 eligible items in stable order (case ID, then file path, then chronological ID). Do not choose examples because they produce a preferred result. If fewer than 12 eligible items exist, use all and record that fact.

Recommended research questions

- RQ1. Can the selected downstream-maintainability result be reproduced from the public data?
- RQ2. Can students reliably identify AI provenance from design quality alone?

You may refine wording during Lab 3 so the RQs match your exact bounded unit of analysis. Keep two RQs unless a third one is clearly necessary and manageable.

Week 5: what to do first

1. Open the paper and linked artefact; locate the paper section/RQ/result that your bounded task corresponds to.
2. Confirm the bounded sample identifiers and enter them in the shared group workbook before running the main analysis.
3. Freeze RQ1/RQ2, method, metric/decision rule, expected result and non-reproduction condition.
4. Run only the minimum scripts/builds needed for the bounded target. Do not recreate a large research infrastructure when published data/output directly supports the target.
5. Keep raw/intermediate output, script/command version and repository/tag references needed to audit your result.

Evidence you must keep

- Exact paper/artefact version or access date and bounded sample IDs.
- Commands/scripts/notebook cells or a clear manual coding rule.
- Raw/intermediate result supporting every quantitative/qualitative claim.
- Any deviation from the paper and the most important threat/limitation.
- Repository commits/tags and tests/verification for any design/refactoring/modernisation work.
- Only brief material AI-use notes when AI changed a decision or result interpretation.

Important scope rule

Keep to the bounded task; do not reproduce the whole paper. Use the smallest public artefact route that still answers the RQs. If a genuine external artefact fails, record the failure and follow the module contingency; do not invent results.